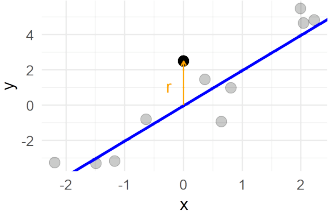
Introduction to Machine Learning

Advanced Risk Minimization
Pseudo-Residuals and Gradient Descent



Learning goals

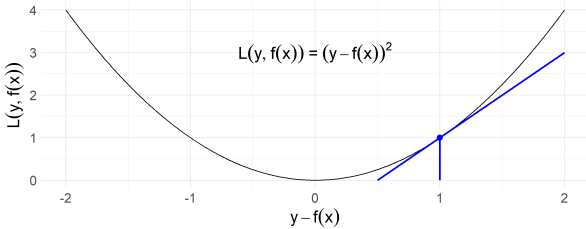
- Know the concept of pseudo-residuals
- Understand the relationship between pseudo-residuals and gradient descent

PSEUDO-RESIDUALS

- In regression, residuals are defined as $r := y - f(\mathbf{x})$.
- We further define **pseudo-residuals** as the negative first derivatives of loss functions w.r.t. $f(\mathbf{x})$

$$\tilde{r} \quad := \quad -\frac{\partial L\left(y, f(\mathbf{x})\right)}{\partial f(\mathbf{x})}.$$

- This definition also holds for score / probability based classifiers.
- Note that $\tilde{r}$ depends on y and $f(\mathbf{x})$ and L.

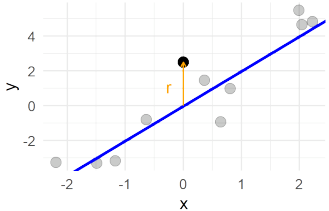


BEST POINT-WISE UPDATE

Assume we have (partially) fitted a model $f(\mathbf{x})$ to data $\mathcal{D}$.

Assume we could update $f(\mathbf{x})$ point-wise as we like. Under squared loss, for a fixed $\mathbf{x} \in \mathcal{X}$, the best point-wise update is the direction of the residual $r = y - f(\mathbf{x})$

$$f(\mathbf{x}) \leftarrow f(\mathbf{x}) + r$$



Caveat: In GB, we don't do point-wise updates but "smoothly distort" f so we generalize

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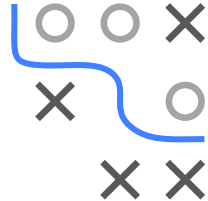

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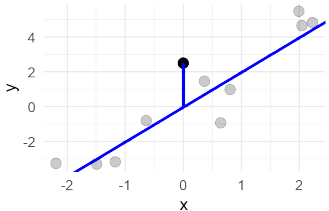
The point-wise error at this specific $\mathbf{x}$ becomes 0.



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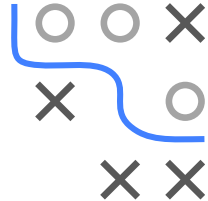

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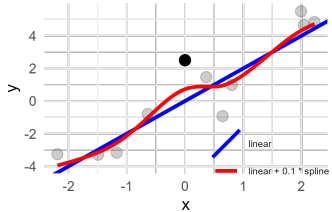

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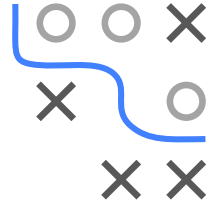
APPROXIMATE BEST POINT-WISE UPDATE



When applying gradient descent (GD) to compute a point-wise update of $f(\mathbf{x})$, we would go a step into the direction of the negative gradient

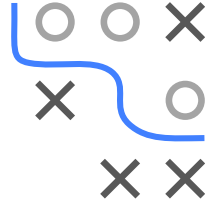
$$f(\mathbf{x}) \leftarrow f(\mathbf{x}) - \frac{\partial L\left(y, f(\mathbf{x})\right)}{\partial f(\mathbf{x})}.$$

which is the direction of the pseudo-residual

$$f(\mathbf{x}) \quad \leftarrow \quad f(\mathbf{x}) + \tilde{r}$$

Iteratively stepping towards the direction of the pseudo-residuals is the underlying idea of gradient boosting, which is a learning algorithm that will be covered in a later chapter.

GD IN ML AND PSEUDO-RESIDUALS



- In GD, we move in the direction of the negative gradient by updating the parameters:

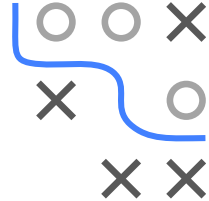
$$\boldsymbol{\theta}^{[t+1]} = \boldsymbol{\theta}^{[t]} - \alpha^{[t]} \cdot \nabla_{\boldsymbol{\theta}} \, \mathcal{R}_{\text{emp}}(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\boldsymbol{\theta}^{[t]}}$$

- This can be seen as approximating the unexplained information (measured by the loss) through a model update.
- Using the chain rule:

$$\begin{aligned} \nabla_{\boldsymbol{\theta}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) &= \sum_{i=1}^{n} \left. \frac{\partial L\left(y^{(i)}, f\right)}{\partial f} \right|_{f=f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)} \cdot \nabla_{\boldsymbol{\theta}} f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right) \\ &= -\sum_{i=1}^{n} \tilde{r}^{(i)} \cdot \nabla_{\boldsymbol{\theta}} f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right). \end{aligned}$$

- Hence the update is determined by a loss-optimal directional change of the model output and a loss-independent derivate of f. This is a very flexible, nearly loss-independent formulation of GD.

PSEUDO-RESIDUALS FOR COMMON LOSSES



Loss	**Domain of y**	**Pseudo Residual $\tilde{r}$**
Squared loss	$y \in \mathbb{R}$	$y - f(\mathbf{x})$
Bernoulli loss	$y \in \{0, 1\}$	$y - s(f(\mathbf{x})) = y - \pi(\mathbf{x})$
Multinomial loss	$y \in \{1, \dots, g\}$	$\mathbb{1}_{\{y=k\}} - \pi_k(\mathbf{x})$
Exponential loss	$y \in \{-1, 1\}$	$y \exp(-yf(\mathbf{x}))$

NB: $\pi(\mathbf{x}) = s(f(\mathbf{x})) = \frac{\exp(f(\mathbf{x}))}{1+\exp(f(\mathbf{x}))}$ is the (sigmoidal) logistic function, and $\pi_k(\mathbf{x})$ is its multi-class generalization, the softmax.